Experiment 2

1. Relational Schema Design

The following table outlines the mapping of the E-Commerce ER Diagram to the relational database schema, highlighting the Primary Keys (PK) and Foreign Keys (FK) for each table.

Table	Primary Key	Important Foreign Keys
CUSTOMER	CustomerID	-
CUSTOMER_PHONE	(CustomerID, Phone)	CustomerID → CUSTOMER
CUSTOMER_ALT_PHONE	(CustomerID, AltPhone)	CustomerID → CUSTOMER
SELLER	SellerID	-
SELLER_PHONE	(SellerID, Phone)	SellerID → SELLER
ADDRESS	AddressID	CustomerID → CUSTOMER
ADDRESS_MOBILE	(AddressID, MobileNo)	AddressID → ADDRESS
CATEGORY	CategoryID	ParentCategoryID → CATEGORY
PRODUCT	ProductID	SellerID → SELLER, CategoryID → CATEGORY
PRODUCT_IMAGE	(ProductID, ImageURL)	ProductID → PRODUCT

Table	Primary Key	Important Foreign Keys
ELECTRONICS	ProductID	ProductID → PRODUCT
CLOTHING	ProductID	ProductID → PRODUCT
BOOKS	ProductID	ProductID → PRODUCT
ORDERS	OrderID	CustomerID → CUSTOMER, AddressID → ADDRESS
ORDER_COUPON	(OrderID, CouponCode)	OrderID → ORDERS
ORDERITEM	(OrderID, ItemNo)	OrderID → ORDERS, ProductID → PRODUCT
PAYMENT	PaymentID	OrderID → ORDERS
DELIVERY	DeliveryID	OrderID → ORDERS, AddressID → ADDRESS

2. Data Definition Language (DDL) Commands

The raw input statements were reviewed and corrected for standard SQL syntactical integrity.

```
CREATE DATABASE ECOMMERCE;
USE ECOMMERCE;

CREATE TABLE CUSTOMER (
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Password VARCHAR(100) NOT NULL,
```

```
    DateOfBirth DATE NOT NULL
);

CREATE TABLE CUSTOMER_PHONE (
    CustomerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY (CustomerID, Phone),
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID) ON DELETE CASCADE
);

CREATE TABLE CUSTOMER_ALT_PHONE (
    CustomerID INT,
    AltPhone VARCHAR(15),
    PRIMARY KEY (CustomerID, AltPhone),
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID) ON DELETE CASCADE
);

CREATE TABLE SELLER (
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    GSTIN VARCHAR(20) UNIQUE NOT NULL,
    AccountNo VARCHAR(20),
    IFSCCode VARCHAR(15),
    BankName VARCHAR(50)
);

CREATE TABLE SELLER_PHONE (
    SellerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY (SellerID, Phone),
    FOREIGN KEY (SellerID) REFERENCES SELLER(SellerID) ON DELETE CASCADE
);

CREATE TABLE CATEGORY (
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(50) NOT NULL,
    Description VARCHAR(100),
    ParentCategoryID INT,
```

```
    FOREIGN KEY (ParentCategoryID) REFERENCES CATEGORY(CategoryID) ON DELETE SET
NULL
);

CREATE TABLE PRODUCT (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50) NOT NULL,
    Description VARCHAR(100),
    Price DECIMAL(10,2) NOT NULL,
    StockQty INT NOT NULL,
    Weight DECIMAL(8,2),
    SellerID INT NOT NULL,
    CategoryID INT,
    FOREIGN KEY (SellerID) REFERENCES SELLER(SellerID) ON DELETE CASCADE,
    FOREIGN KEY (CategoryID) REFERENCES CATEGORY(CategoryID) ON DELETE SET NULL
);

CREATE TABLE PRODUCT_IMAGE (
    ProductID INT,
    Image VARCHAR(200),
    PRIMARY KEY (ProductID, Image),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE ELECTRONICS (
    ProductID INT PRIMARY KEY,
    Brand VARCHAR(50),
    Warranty VARCHAR(30),
    PowerRating VARCHAR(30),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE CLOTHING (
    ProductID INT PRIMARY KEY,
    Size VARCHAR(20),
    Colour VARCHAR(20),
    Material VARCHAR(30),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);
```

```
CREATE TABLE BOOKS (
    ProductID INT PRIMARY KEY,
    ISBN VARCHAR(20) UNIQUE,
    Author VARCHAR(50),
    Publisher VARCHAR(50),
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE ADDRESS (
    AddressID INT PRIMARY KEY,
    DoorNo VARCHAR(10),
    Street VARCHAR(50),
    Area VARCHAR(50),
    City VARCHAR(30),
    State VARCHAR(30),
    Pincode VARCHAR(10) NOT NULL,
    AddressType VARCHAR(20),
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID) ON DELETE CASCADE
);

CREATE TABLE ADDRESS_MOBILE (
    AddressID INT,
    MobileNo VARCHAR(15),
    PRIMARY KEY (AddressID, MobileNo),
    FOREIGN KEY (AddressID) REFERENCES ADDRESS(AddressID) ON DELETE CASCADE
);

CREATE TABLE ORDERS (
    OrderID INT PRIMARY KEY,
    OrderDate DATE NOT NULL,
    OrderStatus VARCHAR(20) NOT NULL,
    TotalAmount DECIMAL(10,2) NOT NULL,
    CustomerID INT NOT NULL,
    AddressID INT,
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID) ON DELETE
CASCADE,
    FOREIGN KEY (AddressID) REFERENCES ADDRESS(AddressID) ON DELETE SET NULL
```

```
);

CREATE TABLE ORDER_COUPON (
    OrderID INT,
    CouponCode VARCHAR(30),
    PRIMARY KEY (OrderID, CouponCode),
    FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID) ON DELETE CASCADE
);

CREATE TABLE ORDERITEM (
    OrderID INT,
    ItemNo INT,
    ProductID INT NOT NULL,
    ItemPrice DECIMAL(10,2) NOT NULL,
    Quantity INT NOT NULL,
    Discount DECIMAL(10,2),
    TaxAmount DECIMAL(10,2),
    PRIMARY KEY (OrderID, ItemNo),
    FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID) ON DELETE CASCADE,
    FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID) ON DELETE CASCADE
);

CREATE TABLE PAYMENT (
    PaymentID INT PRIMARY KEY,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentMethod VARCHAR(30) NOT NULL,
    TransactionID VARCHAR(30) UNIQUE,
    PaymentStatus VARCHAR(20) NOT NULL,
    OrderID INT UNIQUE NOT NULL,
    FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID) ON DELETE CASCADE
);

CREATE TABLE DELIVERY (
    DeliveryID INT PRIMARY KEY,
    DeliveryDate DATE,
    DeliveryStatus VARCHAR(20) NOT NULL,
    TrackingID VARCHAR(30) UNIQUE,
    DeliveryPartner VARCHAR(50),
```

```
    ShippingCharge DECIMAL(8,2),
    OrderID INT UNIQUE NOT NULL,
    AddressID INT,
    FOREIGN KEY (OrderID) REFERENCES ORDERS(OrderID) ON DELETE CASCADE,
    FOREIGN KEY (AddressID) REFERENCES ADDRESS(AddressID) ON DELETE SET NULL
);
```

3. Data Manipulation Language (DML) - Data Insertion

```
INSERT INTO CUSTOMER VALUES (1, 'Unnati', 'unnati10mishra@gmail.com', 'pass123',
'2006-01-10');
INSERT INTO CUSTOMER_PHONE VALUES (1, '9214293443');
INSERT INTO CUSTOMER_ALT_PHONE VALUES (1, '9123456780');

INSERT INTO SELLER VALUES (101, 'TechStore', 'tech@gmail.com', 'GST123', '123456',
'SBIN001', 'SBI');
INSERT INTO SELLER_PHONE VALUES (101, '9123456789');

INSERT INTO CATEGORY VALUES (201, 'Electronics', 'Electronic items', NULL);
INSERT INTO CATEGORY VALUES (202, 'Mobiles', 'Mobile phones', 201);

INSERT INTO PRODUCT VALUES (301, 'Laptop', 'Gaming Laptop', 75000, 10, 2.5, 101, 201);
INSERT INTO ELECTRONICS VALUES (301, 'ASUS', '2 Years', '180W');
INSERT INTO PRODUCT_IMAGE VALUES (301, 'laptop.jpg');

INSERT INTO ADDRESS VALUES (401, '105', 'Krishna Nagar', 'SBI Bankr', 'Unnao', 'Uttar Pradesh',
'209801', 'Home', 1);
INSERT INTO ADDRESS_MOBILE VALUES (401, '7985131870');

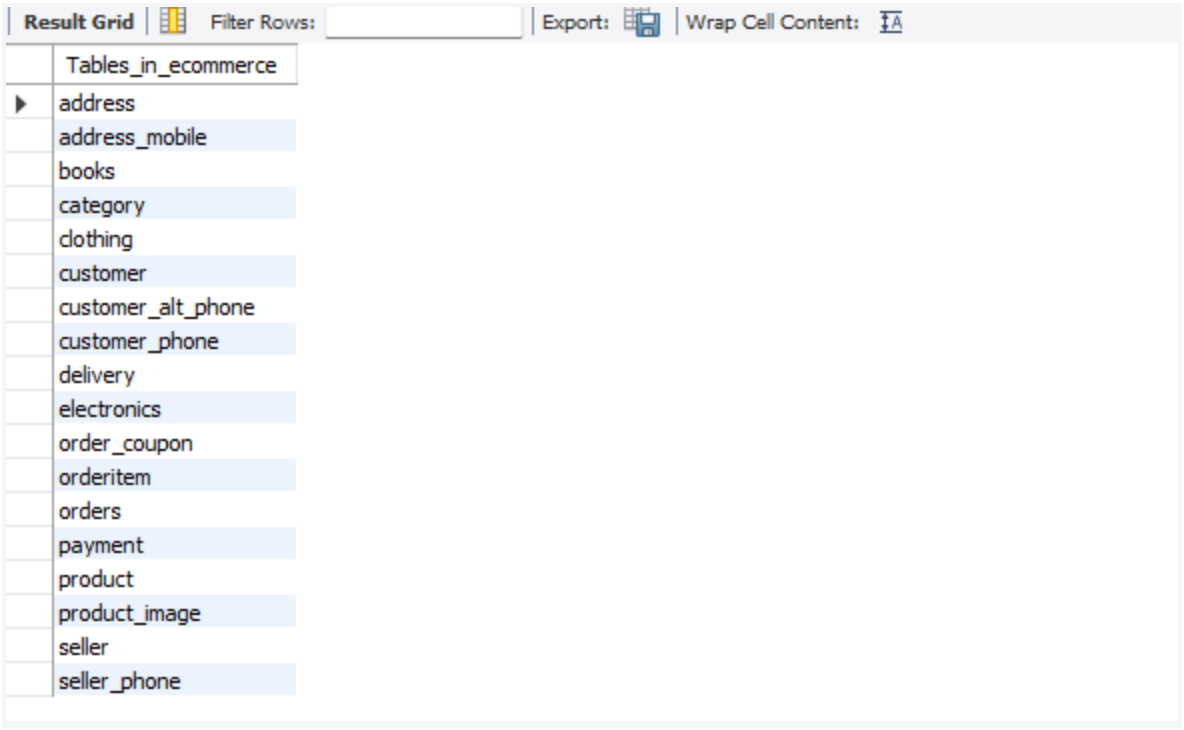
INSERT INTO ORDERS VALUES (501, '2026-08-14', 'Placed', 75000, 1, 401);
INSERT INTO ORDERITEM VALUES (501, 1, 301, 75000, 1, 1000, 9000);
INSERT INTO ORDER_COUPON VALUES (501, 'WELCOME10');

INSERT INTO PAYMENT VALUES (601, '2026-08-14', 74000, 'UPI', 'TXN123', 'Paid', 501);
INSERT INTO DELIVERY VALUES (701, NULL, 'Processing', 'TRK123', 'Delhivery', 100, 501, 401);
```

4. Output Demonstrations and Integrity Checks

Output 1 - SHOW TABLES

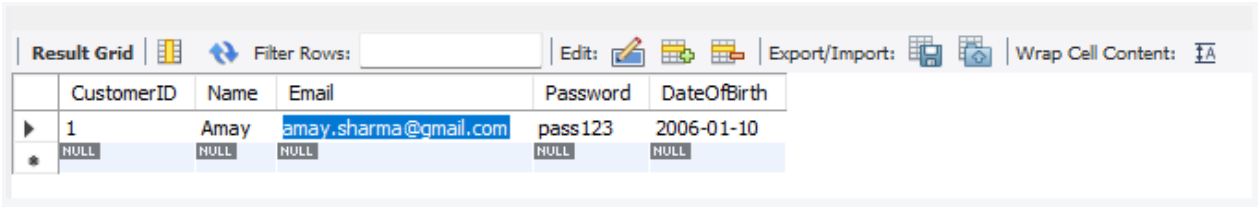
After executing the schema creation scripts, verifying all tables are active:



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

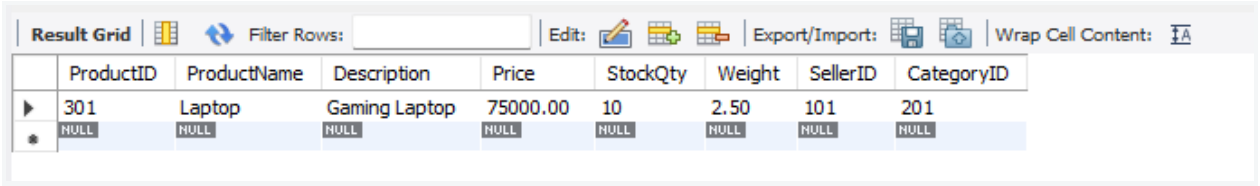
Tables_in_ecommerce
address
address_mobile
books
category
clothing
customer
customer_alt_phone
customer_phone
delivery
electronics
order_coupon
orderitem
orders
payment
product
product_image
seller
seller_phone

Output 2 - Verifying Data Insertion



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

CustomerID	Name	Email	Password	DateOfBirth
1	Amay	amay.sharma@gmail.com	pass123	2006-01-10
NULL	NULL	NULL	NULL	NULL



Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ProductID	ProductName	Description	Price	StockQty	Weight	SellerID	CategoryID
301	Laptop	Gaming Laptop	75000.00	10	2.50	101	201
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

OrderID	OrderDate	OrderStatus	TotalAmount	CustomerID	AddressID
501	2026-08-14	Placed	75000.00	1	401
NULL	NULL	NULL	NULL	NULL	NULL

Output 3 - Referential Integrity Violation

47 14:19:20 INSERT INTO ORDERS VALUES (502, '2026-08-14', 'Placed', 50000, 999, 401) Error Code: 1452. Cannot add or update a child row: a foreign key constraint fails (`ecommerce`.`orders`, CONSTRAINT `orders_ibfk_1` FOREIGN KEY (... 0.031 sec

INSERT INTO ORDERS VALUES (502, '2026-08-14', 'Placed', 50000, 999, 401);

ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails
Reason: CustomerID 999 does not exist in CUSTOMER.
Satisfies "demonstrate referential integrity violations."

Output 4 - ON DELETE SET NULL Demonstration

-- Before deletion:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

ProductID	ProductName	CategoryID
301	Laptop	201
NULL	NULL	NULL

-- After deletion:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

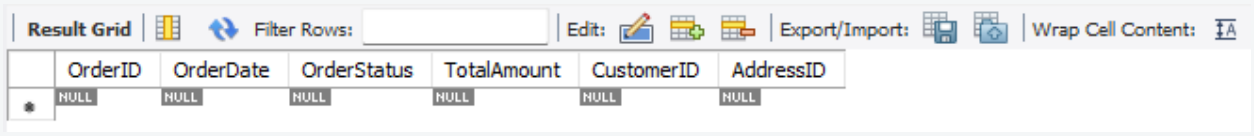
ProductID	ProductName	CategoryID
301	Laptop	NULL
NULL	NULL	NULL

Output 5 - ON DELETE CASCADE Demonstration

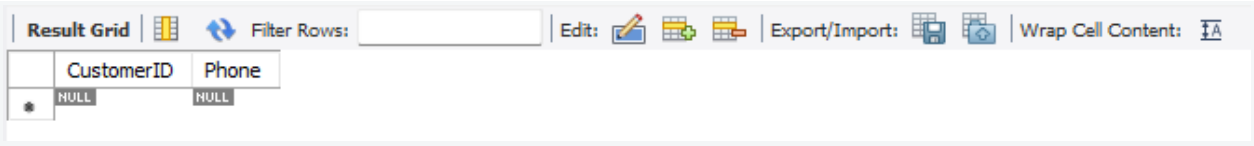
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

OrderID	OrderDate	OrderStatus	TotalAmount	CustomerID	AddressID
501	2026-08-14	Placed	75000.00	1	401
NULL	NULL	NULL	NULL	NULL	NULL

-- After deleting:



SELECT * FROM CUSTOMER_PHONE;



This demonstrates that deleting the parent customer automatically deleted the related records (Orders and Phone details).

ON DELETE CASCADE demonstrated successfully.